Software Requirements Specification

for

<Campus Lost-and-Found Collection, Matching, and Retrieval System>

Version 1.0

Prepared by team 10

KAIST CS30500

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

- This document specifies version 1.0 of the Campus Lost-and-Found Collection, Matching, and Retrieval System ("the system"). The system lets people on a university campus report lost and found items, links lost reports to found items, and manages the return of each item to its verified owner through the campus lost-and-found office. This SRS covers the whole system.

1.2 Document Conventions

- This document follows the Software Requirements Specification template by Karl Wiegers.
- "Shall" and "must" both mark a requirement that the delivered system has to satisfy.
- Each functional requirement in Section 4 has a unique tag of the form REQ-x.y, where x is the number of the system feature (4.x).
- Each system feature has one priority (High, Medium, or Low). Every requirement under a feature inherits that priority.
- Item states are capitalized (Registered, Stored, Recovered). They are defined in Appendix B.
- TBD-n marks information that is not yet decided. All TBD items are collected in Appendix C.

1.3 Intended Audience and Reading Suggestions

- Developer team (the team that will build the system): read Section 2 for the context, then Section 4 feature by feature, then Sections 3 and 5 for the interface and quality requirements that apply to those features.
- Client team (the authors of this document): use Section 4 to check that each delivered feature behaves as specified.

Section 1 introduces the product. Section 2 describes its context, users, and constraints. Section 3 specifies external interfaces. Section 4 specifies the functional requirements, organized by system feature. Section 5 specifies non-functional requirements and business rules.

1.4 Product Scope

When an item is lost on campus today, there is no single place to report it or to look for it. Three problems follow.

- Reports are scattered. Found items are announced in group chats, on notice boards, or left at the nearest office. The owner has to ask in several places and cannot tell whether the item was already returned.
- Lost and found reports are never linked. A person who lost an item and a person who found it may both report it, but nothing connects the two reports.
- Ownership is not checked. Anyone can say "that is mine". There is no agreed way to verify the claim before the item is handed over.

The system gives the campus one place for all three steps: collection (lost and found reports, with found items brought to an official storage location), matching (the system proposes which found

items may belong to which lost report), and retrieval (an administrator verifies ownership before the item is released). Its objectives are:

- A person who lost an item can find out in one search whether it was found.
- A found item is released only to a person whose ownership an administrator has verified.
- Every item has a recorded state from registration to return or disposal, so nobody keeps searching for an item that was already returned.

1.4.1 In scope

- Reporting lost items and found items on campus, by campus members.
- Searching reported items by category, location, and date.
- Automatic generation of match candidates between lost reports and found items.
- Retrieval requests, ownership verification by an administrator, and pickup at an official location.
- Special handling of hazardous items.
- Administrator management of stored items, including unclaimed items.

1.4.2 Out of scope

- Items lost or found outside the campus.
- Direct contact or chat between the finder and the owner. Items are returned only through the official location.
- Delivery or shipping of items.
- Rewards or payments.
- Integration with the police lost-and-found portal (LOST112) or other external lost-and-found services.

1.5 References

- Karl Wiegers, Software Requirements Specification template. The template whose structure this document follows.
- LOST112, the lost-and-found portal of the Korean National Police Agency, https://www.lost112.go.kr. Referred to in Section 1.4 (Out of scope).

2. Overall Description

2.1 Product Perspective

- This system is a new web app for KAIST. KAIST students, faculty, and staff can use it to report lost and found items.

- A person who finds an item can enter its name, where it was found, and where it is kept. A person who loses an item can search for it using its name, place, and other details.

- The system has a website, a web server, a database, and storage for photos. Android and iOS apps will show the same website through a WebView.

2.2 Product Functions

The system will have the following main functions:

2.2.1. Sign up
- Users can sign up with a KAIST email address ending in `@kaist.ac.kr`. They must confirm that the email address is theirs before using the app.
- Users without a KAIST email can upload a document. It must show that they study or work at KAIST. The administrator checks the document and decides whether to approve the account.

2.2.2. Report a lost item
- Users can enter the item's name, type, color, and description. They can also enter when they lost it and where they last saw it. They can add a photo if they have one.

2.2.3. Report a found item
- Users can enter the item's name, type, color, and description. They can also enter when and where they found it and add photos.
−After submitting a found-item report, users must bring the found item to the designated official storage location.

2.2.4. Search for an item
- Users can search found-item reports by keyword. They can also use filters for item type, color, date, and place.
- Users can read the details and look at photos.

2.2.5. View the pickup place
- The person who lost it can go to the archive to retrieve the lost item.
- The administrator verifies ownership before the item is released.

2.2.6. Update and close reports
- Users can edit their own reports. A finder can change where an item is kept or mark it as picked up.
- Users can close their lost-item reports when they get their items back.

2.2.7. Manage the app
- The administrator can approve accounts, manage user access, and remove reports when needed. The administrator can also check reported problems and manage saved data.

2.3 User Classes and Characteristics

2.3.1 Campus Members
- KAIST undergraduate students, graduate students, faculty, staff, and temporary workers can join. All approved members can use the same main functions. They can report items, search reports, view pickup places, and edit their own reports.
- Members without a KAIST email need approval from the administrator. After approval, they can use the same functions as other members.

- Most members will use the app only when they lose or find an item. The app must be easy for first-time users.

2.3.2 Administrator

- The administrator is the person who manages the app. The project owner will take this role.
- The administrator checks sign-up documents, approves accounts, manages access, and checks reported problems. The administrator can view private information needed for these tasks.

2.4 Operating Environment

- Users can open the website on a computer, tablet, or smartphone. They need an Internet connection.
- The website will use CSS to fit different screen sizes. Text, photos, forms, and buttons must be easy to use on each device.
- The website will support Chrome, Edge, Firefox, and Safari.
- The Android and iOS apps will use a WebView to display the website within the app. The project plans to publish these apps on Google Play and the Apple App Store.
- A web server will run the system. A database will save accounts, reports, and button records. File storage will save photos and sign-up documents.
- The team will choose the server software, database, and hosting service before building the system.

2.5 Design and Implementation Constraints

- Users must confirm their KAIST email or get approval from the admin.
- The app must protect data when it is sent and saved. Saved personal data and sign-up documents must be encrypted to keep them private.
- Passwords must be saved in a protected form that does not show the original password.
- Regular users must not see other users' private account information or button records.
- The team must decide how long to keep reports, photos, sign-up documents, and button records before release. Data must be deleted when that time ends.

2.6 User Documentation

The app will include these guides:
- Getting started: How to sign up, confirm an email address, and ask for account approval.
- Using the app: How to report an item, add photos, search, view pickup places, and update reports.
- Help page: Answers to common questions and steps for reporting a problem or contacting the admin.

The guides will be available in the website and mobile apps. They will use short sentences, examples, and simple steps.

2.7 Assumptions and Dependencies

We expect that:
- Users have Internet access and can use a website or mobile app.
- Users enter correct information.
- Users take only items that belong to them.
- Finders choose places that campus members can reach.
- Finders update their reports when the pickup place changes.
- Users without a KAIST email can provide a document that shows they study or work at KAIST.
- The administrator has time to check new accounts and report problems.

The system needs a working web server, database, and file storage.
The system needs an email service to send verification emails to users' KAIST email address.
Publishing the mobile apps needs developer accounts and approval from the app stores.
Search results depend on the information in reports. Wrong names, dates, or places can make an item harder to find.
Button records show who viewed the pickup place. They can help the administrator check later problems. However, the app cannot directly check who actually took the item.

3. External Interface Requirements

3.1 User Interfaces

3.1.1 User Types
The system shall provide user interfaces for two types of users: Users and Administrators.
- User: A general user who can interact with the system as either a lost-item owner or a person who reports a found item. Depending on the situation, a user may report a lost item, report a found item, search for items, and perform other functions provided to general users.

- Administrator: A person responsible for managing the system and lost-and-found items. The administrator can manage reported items, review reports, manage the matching process, and handle the return of items to their owners.

3.1.2 User Interface Screens
- Login Screen: The Login Screen is the entry point to the system for users and administrators. It allows users to enter their authentication information and access the system. The system identifies the user's role and provides access to the appropriate functions. If the provided authentication information is invalid, the screen shall display an appropriate error message.

- Main Screen: The main screen shall provide an overview of the primary functions available to the user. It shall allow users to navigate to functions such as reporting a lost or found item, searching for items, and viewing relevant information. The functions displayed on the screen may differ depending on the user's role.

- Lost Item Report Screen: The Lost Item Report Screen allows users to provide information about an item they have lost. The screen shall provide input fields for information such as the item name, category, location, date, and description. It also allows users to upload an image of the lost item. The screen shall provide a way for users to submit the completed report and receive feedback

about the submission. The system shall require users to enter all required information before submitting the report. The screen shall display an appropriate confirmation message when the report is successfully submitted.

- Found Item Report Screen: The Found Item Report Screen allows users to provide information about an item they have found. The screen shall provide input fields for information such as the item name, category, location, date, and description. It also allows users to upload an image of the found item. The system shall require users to enter all required information before submitting the report. The screen shall display an appropriate confirmation message when the report is successfully submitted.

- Search Screen: The Search Screen shall allow users to search for lost and found items registered in the system. Users shall be able to search using relevant information such as category, location, and date. The screen shall display the items that match the selected search criteria and allow users to select an item to view its detailed information. The search results shall provide basic information about each item.

- Item Details Screen: The Item Details Screen will display detailed information about a selected lost or found item. The screen will provide information such as name, category, location, date, description, image, depending on the information available. The screen shall provide relevant actions according to the item's status and the user's role, such as requesting the return of a found item.

- Administrator Screen: The Administrator Screen will provide administrators with the functions required to manage the lost-and-found system. It must allow administrators to review submitted reports, view and update item information and status, manage matching results, and process return requests. Administrators shall also be able to delete information about lost items that have remained in the system for a long period of time and ensure that matched lost-and-found items are no longer visible on the user screens. The Administrator Screen will display the current status of reported items and allow administrators to update the status when necessary.

3.1.3 User Interface Requirements

- Users shall be able to navigate between the main functions of the system through the Main Screen.
- The system shall require users to provide all required information when submitting a lost or found item report. If required information is missing or invalid, the system should print an appropriate error message.
- The system must allow users to upload images when reporting found items and provide appropriate feedback if the uploaded file is invalid.
- The system shall provide appropriate feedback for actions such as submitting reports, requesting the return of an item, and processing matching results.
- The interface shall have access to administrative functions.

3.2 Hardware Interfaces

3.2.1 Desktop Computers and Laptops

- The system shall support access through desktop computers and laptops. Users shall be able to access the system using a keyboard and mouse to enter information, navigate between screens, search for lost and found items, and submit reports. Users also will be able to upload images of items stored on the device. The system shall support commonly used desktop and laptop operating systems and shall not require specialized hardware for basic system functions.

3.2.2 Smartphones and Tablets

- The system shall support access through smartphones and tablets. Users shall be able to enter information, navigate between screens, search for lost and found items, and submit reports using touchscreens. Users also will be able to upload images of items stored on the device. The system shall provide an interface that is appropriately displayed according to the screen size of the smartphone or tablet. The system shall not require specialized hardware for basic system functions.

3.2.3 Camera

- The system shall support the use of a device camera when users need to capture images of items. Users shall be able to take a photograph using the camera and use the captured image when submitting a lost-and-found item report.

3.2.4 Server and Storage

- The server shall provide the computing and storage resources required to operate the system. It shall process user requests and store system data, including lost and found item information, uploaded images, matching results, and return request information.

3.2.5 Network Devices

- The system shall use the network connection provided by the user's device to communicate with the server. The network connection shall support the transmission and reception of data, including item reports, search results, and return request information.

3.3 Software Interfaces

3.3.1 Database

- The system shall communicate with the database to store, retrieve, and update lost-and-found item information.

- The database shall store lost-and-found items as separate data classes.

- The Lost Item class shall store information about each reported lost item, including the item name, category, location where the item was lost, date and time when the item was lost, description, image, and current status.

- The Found Item class shall store information about each reported found item, including the item name, category, location where the item was found, date and time when the item was found, description, image, and current status.

- The database shall also store information required to associate lost-and-found items with their respective users and to manage matching results and return requests.

- The system shall retrieve the relevant item information from the database when users perform searches and shall update the corresponding data when users or administrators modify item information or status.

- The system shall ensure that only authorized users and administrators can access or modify the data stored in the database.

3.4 Communications Interfaces

3.4.1 Client-Server Communication

- The system will be able to use HTTPS to communicate between the user's devices and the system server and protect data transmitted between the client and the server from unauthorized access or modification.
- The system shall support the transmission and reception of data required for system functions, including user requests, lost-and-found item reports, search requests, search results, images, and return requests.
- The system shall provide an appropriate response to the client when a communication error or data transmission failure occurs.

3.4.2 File Transfer Communication

- The system shall support the transmission of images between the client and the server for lost-and-found item reports.
- The system shall transmit uploaded images securely between the client and the server using HTTPS.
- The system shall provide appropriate feedback when an image upload fails due to an invalid file format, file size, or every communication error.

4. System Features

The features are organized in the order an item moves through the system: access, reporting, search, matching, retrieval, and administration. Section 3.1 describes the screens; this section specifies what the system must do behind them.
The states of a found item (Registered, Stored, Recovery requested, Ownership verified, Recovered, Disposed) and of a lost report (Open, Closed, Withdrawn) are defined in Appendix B and follow the lifecycle in Section 5.5.2.

4.1 Login and User Access

4.1.1 Description and Priority

Campus members log in with their verified KAIST email address and password and receive the functions of their role.

4.1.2 Stimulus/Response Sequences

1. A person opens the system. The system shows the Login Screen.

2. The person logs in with a KAIST email address and password. The system identifies the person's role and shows the Main Screen with the functions of that role.
2a. The authentication information is invalid. The system shows an error message and stays on the Login Screen.

4.1.3 Functional Requirements

REQ-1.1: The system shall authenticate users using their verified KAIST email address and password after email verification.
REQ-1.2: The system shall assign each authenticated account one role: User or Administrator (see TBD-1).
REQ-1.3: The system shall show a person only the functions that the person's role permits.
REQ-1.4: If authentication fails, the system shall show an error message and shall not grant access.

4.2 Lost Item Report

4.2.1 Description and Priority

A person who lost an item on campus records what was lost, where, and when, so that it can be matched with found items. Priority: High.

4.2.2 Stimulus/Response Sequences

1. The user selects "report a lost item" on the Main Screen. The system shows the Lost Item Report Screen.
2. The user enters the item name, category, location, date, and description, optionally uploads an image, and submits.
3. The system saves the report as Open and confirms the submission.
4. The system shows the found items that are match candidates for the report (4.5).
2a. A required field is missing or invalid. The system does not save the report and shows which field must be corrected.
4a. There is no candidate. The system says so and tells the user that a notification will be sent when a matching item is stored.

4.2.3 Functional Requirements

REQ-2.1: The system shall let a User file a lost report with: item name, category, location where the item may have been lost, date of loss, and description.
REQ-2.2: The system shall let the reporter attach an image to a lost report.
REQ-2.3: The system shall refuse a lost report with a missing required field or a date in the future, and shall show which field is wrong.
REQ-2.4: The system shall save an accepted lost report in state Open and show the reporter a confirmation.
REQ-2.5: The system shall let a User view the list and current state of the User's own reports.
REQ-2.6: The system shall let a User edit or withdraw the User's own lost report while it is Open.
REQ-2.7: When an item is recovered through a retrieval request linked to a lost report, the system shall set that lost report to Closed.

4.3 Found Item Reporting and Intake

4.3.1 Description and Priority

A person who found an item reports it and brings it to the official drop-off location. An administrator inspects the item, records how it can be identified, and stores it. Priority: High.

4.3.2 Stimulus/Response Sequences

1. The finder selects "report a found item". The system shows the Found Item Report Screen.

2. The finder enters the item name, category, location, date, and description, uploads an image if available, and submits.

3. The system saves the report as Registered and shows the finder where to bring the item.

4. The finder brings the item to the official drop-off location.

5. An administrator inspects the item, records its storage location and at least one confidential characteristic, and confirms the report.

6. The system sets the item to Stored and searches for match candidates (4.5).

2a. A required field is missing or invalid, or the uploaded file is not a valid image. The system does not save the report and shows what must be corrected.

2b. A found report with the same category, location, and date already exists. The system warns the finder of a possible duplicate and lets the finder continue or cancel.

5a. The item is hazardous. The administrator marks it for special handling (4.8).

4.3.3 Functional Requirements

REQ-3.1: The system shall let a User file a found report with: item name, category, location found, date found, and description.
REQ-3.2: The system shall let the finder upload an image of the item, and shall show an error if the uploaded file is not a valid image.
REQ-3.3: The system shall refuse a found report with a missing required field or a date in the future, and shall show which field is wrong.
REQ-3.4: If a found report with the same category, location, and date already exists, the system shall warn the finder of a possible duplicate before saving.
REQ-3.5: The system shall save an accepted found report in state Registered and show the finder the official drop-off location.
REQ-3.6: The system shall let an administrator confirm a Registered found report after inspecting the item, by recording its storage location and at least one confidential characteristic. The item then becomes Stored.
REQ-3.7: The system shall show the confidential characteristics and the storage location of an item only to administrators.

REQ-3.8: The system shall let an administrator correct the information of any found report.

4.4 Search and Item Details

4.4.1 Description and Priority

Any user can look through reported items without filing a report first. Priority: High.

4.4.2 Stimulus/Response Sequences

1. The user opens the Search Screen and chooses a category, a location, a date, or a combination.

2. The system lists the items that fit all chosen criteria.

3. The user selects an item. The system shows the Item Details Screen.

2a. No item fits. The system says that no item was found.

4.4.3 Functional Requirements

REQ-4.1: The system shall let a User search lost reports and found items by category, location, and date, alone or in combination.
REQ-4.2: The system shall include in search results only lost reports that are Open and found items that are Stored or Recovery requested (see TBD-2).
REQ-4.3: The system shall show, for a selected item, its name, category, location, date, description, and image, as far as each is available.
REQ-4.4: The system shall not show an item's confidential characteristics, its storage location, or any personal information of its reporter on the Item Details Screen.
REQ-4.5: If no item fits the criteria, the system shall say so.

4.5 Matching

4.5.1 Description and Priority

The system proposes which found items may belong to which lost report, so that an owner does not have to read every report. A match is only a suggestion; it never decides ownership. Priority: High.

4.5.2 Stimulus/Response Sequences

1. A lost report is filed, or a found item becomes Stored.

2. The system compares it with the existing reports of the other kind and creates match candidates.

3. The system notifies the reporter of each lost report that has a new candidate.

4. The reporter opens the candidate and reads its item details.

2a. No report fits. The system creates no candidate and notifies nobody.

4a. The reporter recognizes the item. The flow continues with a retrieval request (4.6).

4.5.3 Functional Requirements

REQ-5.1: When a lost report is filed or a found item becomes Stored, the system shall create a match candidate for every pair of an Open lost report and a Stored found item that have the same category and a found date not earlier than the date of loss (see TBD-4).

REQ-5.2: The system shall list the candidates of a lost report with items found at the reported location first, and within each group the most recently found first.

REQ-5.3: The system shall notify the reporter of a lost report when a new candidate is created for it.

REQ-5.4: The system shall let an administrator view the candidates of any report and add or remove a candidate manually.

REQ-5.5: The system shall not change the state of an item or approve a retrieval request because of a match candidate.

4.6 Retrieval Request and Ownership Verification

4.6.1 Description and Priority

A person who believes a stored item is theirs submits a retrieval request. An administrator compares the request with the confidential characteristics recorded at intake and approves it, rejects it, or asks for more proof. Priority: High. This is the feature that prevents an item from going to the wrong person.

4.6.2 Stimulus/Response Sequences

1. The requester opens the details of a Stored found item and selects "request retrieval".

2. The requester enters identification information and describes the confidential characteristics of the item, then submits.

3. The system sets the item to Recovery requested and notifies the administrators.

4. An administrator compares the requester's description with the recorded confidential characteristics.

5. The administrator approves the request.

6. The system sets the item to Ownership verified and shows the requester the official retrieval location and the responsible department.

1a. The requester is the person who reported this found item. The system refuses the request.

3a. Another request on the same item is already waiting. The system keeps both and shows them together to the administrator.

5a. The administrator rejects the request. The system notifies the requester with the reason. If no other request is waiting, the item returns to Stored.

5b. The administrator asks for additional proof. The system notifies the requester, who adds information to the same request. The flow returns to step 4.

4.6.3 Functional Requirements

REQ-6.1: The system shall let a User submit a retrieval request on a found item that is Stored or Recovery requested, with identification information and a description of the item's confidential characteristics.
REQ-6.2: The system shall let the requester link the retrieval request to one of the requester's own lost reports.
REQ-6.3: The system shall refuse a retrieval request from the person who filed the found report of that item.
REQ-6.4: When a retrieval request is submitted, the system shall set the item to Recovery requested and notify the administrators.
REQ-6.5: The system shall accept more than one retrieval request on the same item and show all of them to the administrator together.
REQ-6.6: The system shall show the administrator each requester's description next to the confidential characteristics recorded at intake, and let the administrator approve the request, reject it, or ask for additional proof.
REQ-6.7: When additional proof is asked for, the system shall notify the requester and let the requester add information to the same request.
REQ-6.8: When a request is approved, the system shall set the item to Ownership verified, reject every other waiting request on that item, and notify all requesters of the result.
REQ-6.9: The system shall show the official retrieval location and the responsible department only to the requester whose request was approved.
REQ-6.10: When a request is rejected, the system shall notify the requester with the reason, and shall return the item to store if no other request is waiting.
REQ-6.11: The system shall never approve a retrieval request without an administrator's decision.
REQ-6.12: The system shall not show the finder's personal information to a requester, or a requester's personal information to the finder.

4.7 Pickup and Completion

4.7.1 Description and Priority

The verified owner collects the item at the official retrieval location and an administrator records the handover, which closes the item and the linked lost report. Priority: High.

4.7.2 Stimulus/Response Sequences

1. The verified owner comes to the official retrieval location.

2. An administrator hands over the item and records the handover in the system.

3. The system sets the item to Recovered, saves the date and time, closes the linked lost report, and removes both from search results.

2a. The item is not Ownership verified. The system refuses to record the handover.

3a. An administrator later reports that the item went to the wrong person. The system notifies all administrators and blocks every further action on that item until an administrator resolves the case.

4.7.3 Functional Requirements

REQ-7.1: The system shall let an administrator record the handover of an item that is Ownership verified. The item then becomes Recovered.
REQ-7.2: The system shall refuse to record a handover for an item in any other state.
REQ-7.3: The system shall save the date and time of the handover and the administrator who recorded it.
REQ-7.4: The system shall remove a Recovered item and its linked lost report from search results.
REQ-7.5: The system shall let an administrator report a wrong handover on a Recovered item. The system shall then notify all administrators and refuse further actions on that item by non-administrators until an administrator marks the case resolved.

4.8 Hazardous Item Handling

4.8.1 Description and Priority

Items that violate campus regulations or the law, such as weapons, chemicals, flammable substances, or medication, are not returned like ordinary items (Section 5.2.1). Priority: Medium.

4.8.2 Stimulus/Response Sequences

1. During inspection, an administrator finds that an item is hazardous and marks it for special handling.

2. The system hides the item's storage details from everyone except administrators.

3. A retrieval request on the item is approved. The administrator enters a return location chosen for this item.

4. The system shows that location to the approved requester only.

1a. The item must be reported. The administrator records that it was transferred to the authorities. The system sets the item to Disposed and accepts no retrieval request on it.

4.8.3 Functional Requirements

REQ-8.1: The system shall let an administrator mark a found item as requiring special handling, and remove the mark.

REQ-8.2: For an item marked for special handling, the system shall require the administrator to enter a designated return location when approving a retrieval request, and shall show that location only to the approved requester.
REQ-8.3: The system shall let an administrator record that an item was transferred to the authorities. The item then becomes Disposed and the system shall refuse retrieval requests on it.

4.9 Storage Management and Unclaimed Items

4.9.1 Description and Priority

Administrators keep the records of stored items up to date and clear out items and reports that nobody claimed (Section 5.5.5). Priority: Medium.

4.9.2 Stimulus/Response Sequences

1. An administrator opens the Administrator Screen. The system lists all reports and items with their states.

2. The administrator selects a stored item and updates its status record.

3. The system saves the update with the date and time.

1a. An item has been Stored longer than the retention period. The system marks it as unclaimed in the list.

2a. The administrator disposes of or donates an unclaimed item and records it. The system sets the item to Disposed and removes it from search results.

4.9.3 Functional Requirements

REQ-9.1: The system shall let an administrator list all lost reports and found items and filter them by state.
REQ-9.2: The system shall let an administrator add a status record to a stored item, saved with the date and time.
REQ-9.3: The system shall mark a found item as unclaimed when it has been Stored for TBD-4 days without an approved retrieval request.
REQ-9.4: The system shall let an administrator record that an unclaimed item was disposed of or donated. The item then becomes Disposed and is removed from search results.
REQ-9.5: The system shall let an administrator delete a lost report that has been Open for more than TBD-3 days.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

This system must reliably provide functions such as the registration, search, matching, and return requests for lost and found items on campus. Therefore, it must ensure a stable response speed so that users do not experience delays that cause inconvenience in a typical on-campus environment, while also maintaining service quality even when there are a large number of user requests.

5.1.1 Response Time

- When a user requests a search for a lost or found item, the system must query the database and return search results within an average of 3 seconds.

- The system must process data entry requests—such as registering lost or found items and submitting retrieval requests—including data entry and sorting, within an average of 5 seconds.

5.1.2 Concurrent User Handling

- The system must be designed so that the response time for core functions does not exceed the normal range, even when at least 1% of all users are using those functions simultaneously.

- To prevent service interruptions or issues even during periods of high concurrent request volume, the system must process at least 95% of requests within 7 seconds of receipt.

5.1.3 Matching Performance

- When new lost and found items are registered, the system must automatically search for matches in the existing database and, if necessary, be able to manually add matching categories.

- To ensure search performance does not degrade even as the total data volume increases during the process of generating matching records, the system must employ methods such as automatic indexing and search optimization.

5.1.4 Service Operational Availability

- The system must operate normally for at least 95% of the total duration during the academic semester. If unexpected issues arise with the service, they must be resolved—for example, by automatically suspending operations or providing a previous version of the system.

- If system maintenance is required, users must be notified in advance, and the availability period must not fall below the required threshold.

5.2 Safety Requirements

This section defines safety requirements regarding the process for retrieving lost items, the handling of hazardous materials, and the reporting of false information.

5.2.1 Handling of Hazardous Items

- If items that violate campus regulations or laws—such as weapons (e.g., knives), chemicals, flammable substances, or medications—are registered in the system, they must not be handed over directly to the owner as if they were ordinary lost items. Instead, they must be marked as requiring special handling, and only after an administrator has confirmed that there are no issues should the administrator designate a specific location for their return. If any issues arise, the administrator must report them to the relevant authorities.

- Users other than administrators must not be able to access information regarding the retrieval location or storage details of hazardous items.

5.2.2 Safety Precautions During the Lost and Found Retrieval Process

- Upon a retrieval request, the system must direct the user to the designated official retrieval location and the responsible administrative department.

- The system must prevent the disclosure of unnecessary personal information—such as home addresses and phone numbers—between the person who reported the loss and the owner, thereby ensuring that lost items are not handed over directly without going through the system.

- The system must provide the person who reported the loss with information regarding the location of the lost item only after ownership has been clearly verified.

5.2.3 Handling Safety-Related Errors

- If, due to a system error, a recovered item is delivered to the wrong person, or if a lost item is retrieved before recovery approval or ownership verification is complete, an administrator must be notified immediately, and the system must block access by other users to that request until the issue is resolved.

5.3 Security Requirements

Since this system collects and stores accounts for students, faculty, staff, and administrators, as well as information on lost items and personal data, it must provide access authentication, permission management, and personal information protection features.

5.3.1 User Authentication

- The system must require all users to have an authenticated account. Users shall be authenticated using their verified KAIST email address and password.

- The system must implement preventive measures against authentication attacks, such as brute-force attacks, including limiting the number of consecutive login attempts.

5.3.2 Access Permissions

- The system must support, at a minimum, the following role-based access controls:
General User: Registering and searching for lost and found items; modifying personal data; requesting retrieval

- Intermediate Administrator: Inspecting items; approving matches and retrievals; managing data; processing reports

- Administrator: Managing all user permissions; configuring system settings; handling logs and operations

- General users must not have access to administrator privileges, and intermediate administrators assigned specific roles must not be able to use functions beyond their scope of work.

5.3.3 Privacy Protection
- The system must collect personal information—such as names, student ID numbers, email addresses, and phone numbers—only to the minimum extent necessary for the purpose of the service.

- When transmitting data, personally identifiable information must be protected using encrypted communication methods and must not be stored in plain text in the database.

5.3.4 Auditing and Logging
- The system must log key events, including logins, permission changes, post submissions and deletions, approval of recalls, ownership verification, and data modifications.

- The system's chief administrator must review logs as necessary for security and operational purposes and, if any issues are detected, resolve them through appropriate measures such as forced logout or account access blocking.

5.4 Software Quality Attributes

The system must meet quality attributes such as usability, reliability, maintainability, and scalability to ensure long-term operation.

5.4.1 Usability
- The system must provide a clearly identifiable interface that allows new users to utilize its key functions without additional training.

- If required input values are missing, the user must be notified before data processing begins.

- Buttons, icons, and display formats must be consistent across all service screens.

5.4.2 Reliability
- Even if partial failures—such as database errors or temporary network failures—occur in the operational environment, the service must continue to function to the extent possible.

- A periodic data backup policy must be in place to prevent data loss following recovery from a failure.

5.4.3 Maintainability
- The system's key functions must be modifiable and testable independently of one another and must not affect the current system until changes are implemented.

- The source code must include comments on key logic and development details so that maintenance personnel can clearly understand its operation.

5.4.4 Scalability

- The system must be designed to scale—even as the number of users and the volume of data exceed initial operational projections—while minimizing structural changes.

5.5 Business Rules

This system is designed to support the management of lost and found items on campus; it must clearly distinguish between the roles of general users and administrators and define the procedures for handling items.

5.5.1 User Roles
- All users must be assigned access permissions based on their respective roles.

5.5.2 Finding and Registering Lost Items
- When registering a lost or found item, the user must enter minimum information, such as the location and time of discovery.

- If an identical or similar item has already been registered, the user must be notified of the possibility of a duplicate entry.

- The status of a registered item follows the following lifecycle:
Registration – Inspection and Verification – Storage – Matching – Recovery Request – Ownership Verification – Recovery Completed

5.5.3 Verification of Ownership
- To verify ownership, the user must submit the identification information required by the system and confidential characteristics of the item.

- If ownership verification fails, the administrator must either reject the retrieval request or request additional proof.

- If multiple users simultaneously claim ownership of a single item, the administrator must determine the appropriate owner based on the provided evidence.

5.5.4 Matching Rules
- The system may generate matching candidates using database information. However, automatic matching results must be treated solely as matching candidates and not as a final determination of ownership.

- Even if the system’s matching score is high, ownership must not be automatically confirmed without the administrator’s approval.

5.5.5 Storage and Retrieval
- The administrator must periodically record the status of stored items in the system.

- Upon completion of the retrieval process, the date and time of retrieval and related records must be saved so they can be verified in the event of future issues.

- Items for which the owner has not been identified within a specified period shall be disposed of or donated in accordance with separate regulations.

5.5.6 Data Retention
- Even data for which processing has been completed must be retained for a specified period to facilitate dispute resolution.

- Personal information for which the retention period has expired must be deleted or de-identified.

6. Other Requirements

No additional requirements have been identified for version 1.0 of the system.

Appendix A: Glossary

Term	Definition
Confidential characteristic	A feature of a found item that an administrator records at intake and that is never shown to non-administrators. Used to verify ownership.
Match candidate	A pair of a lost report and a found item that the system proposes as possibly the same item. A candidate never decides ownership.
Retrieval request	A request to receive a stored item as its owner.
Official storage location	The campus place where found items are dropped off, stored, and picked up.
Unclaimed item	A stored item whose retention period passed without an approved retrieval request.

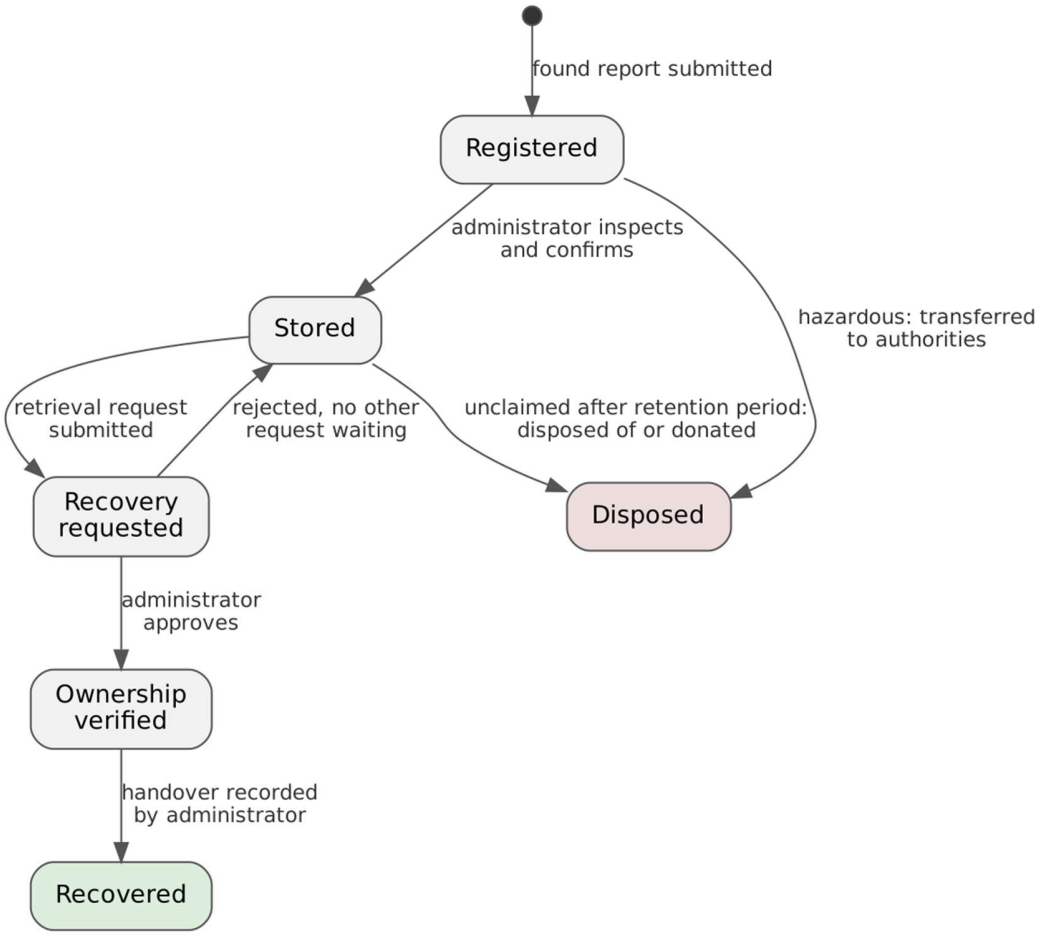
Appendix B: Analysis Models

States of a found item. A found item is always in exactly one of the following states.

State	Meaning
Registered	A found report was submitted; the item has not yet been inspected by an administrator.
Stored	An administrator inspected the item and it is kept at the official storage location.
Recovery requested	At least one retrieval request is waiting for an administrator's decision.
Ownership verified	An administrator approved one retrieval request; the item is waiting for pickup.
Recovered	The item was handed to the verified owner. Final.
Disposed	The item was disposed of, donated, or transferred to the authorities. Final.

States of a lost report. A lost report is Open, Closed (the item was recovered), or Withdrawn (the reporter cancelled it).

State-transition diagram of a found item:



Appendix C: To Be Determined List

TBD-1 Roles: Section 5.3.2 defines an Intermediate Administrator, but Section 3.1 has only User and Administrator. Section 4 assumes one Administrator role.
TBD-2 Whether a found item is visible in search before an administrator has inspected it. Section 4 assumes it becomes visible when stored.
TBD-3 Retention periods in days: how long a found item is stored before it counts as unclaimed, and how long an Open lost report is kept.
TBD-4 Matching criteria: Section 4 uses category and date. Section 5.5.4 mentions a matching score.